From runs to a defensible comparison

Two decisions about how the family of tests is read are taken here, before the results, because either could otherwise be fitted to them

1. Measure in interleaved blocks

20 repeats, with the four variants run back to back inside each, so the interference common to a block is shared rather than assigned to one variant.

2. Summarise without assuming a shape

Median with a nonparametric bootstrap 95% interval, reported in full rather than as a half-width: latency is skewed, and a symmetric width covering the upper bound would fall below zero.

3. Test the comparison, not the overlap

An exact two-sided paired permutation test on the repeat-wise differences, with the allocator held fixed. All 2^20 sign assignments are enumerated, so the p-value is exact.

4. Correct the family

Holm step-down across the 14 comparisons, controlling the family-wise error rate without assuming the tests independent.

Resolving power, quoted rather than left implicit

An exact test on 20 pairs floors at p = 0.0000019, and at 0.000027 after correction.

PRE-COMMITTED

Which comparisons enter the family

Achieved throughput is excluded. Below saturation it is pinned to the offered load by construction, so it cannot express a language effect whatever the languages do. Holm also divides its budget across the whole family, so two comparisons that resolve nothing would make the correction harsher on the ones that can.

Which instrument to believe

When the interval and the p-value disagree, which at these repeat counts they routinely will, the answer turns on whether pairing cancels the interference that metric is exposed to. A run-level summary or a central order statistic is unmoved by a handful of descheduled operations, so a consistent paired difference is real. A high percentile is computed from precisely those operations, so an interval excluding zero is not read as a finding however far from zero it sits.

Applying the second rule to the block that suits the argument and quietly dropping it for the other is the specific failure the pre-commitment exists to prevent.